

```

Private Sub Form_Load()
Text1.Text = ""
Label10.Caption = "DEVICE IS CONNECTED ON PORT NO. :'"
n = Val(InputBox("Check Device Manager Then Type Com Port No. here"))
Text1.Text = n
MSComm1.RThreshold = 3
MSComm1.Settings = "9600,n,8,1"
MSComm1.CommPort = n
MSComm1.PortOpen = True
MSComm1.DTREnable = False
Label10.Caption = "DEVICE IS CONNECTED ON PORT NO. :'"
End Sub

'/'***Arduino Pin No.5---Relay No.1 ON & OFF *****

Private Sub Command1_Click()
Label2.Caption = "ON"
MSComm1.Output = "1"
End Sub

Private Sub Command2_Click()
Label2.Caption = "OFF"
MSComm1.Output = "0"
End Sub

'/'***Arduino Pin No.6---Relay No.2 ON & OFF *****
Private Sub Command3_Click()
Label3.Caption = "ON"
MSComm1.Output = "3"
End Sub

Private Sub Command4_Click()
Label3.Caption = "OFF"
MSComm1.Output = "2"
End Sub

'/'***Arduino Pin No.7---Relay No.3 ON & OFF *****
Private Sub Command5_Click()
Label4.Caption = "ON"

```

Fig. 5: Visual Basic GUI code

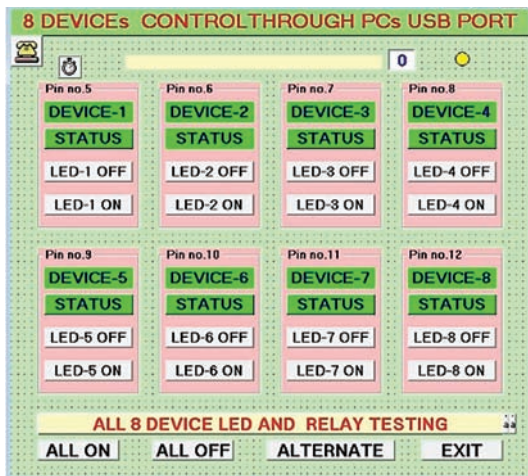


Fig. 6: Visual Basic GUI for appliance control

4. Run the Arduino IDE program on the PC.

5. If not using an Arduino Uno board, use a USB-to-TTL serial converter card to program the ATmega328P. Connect the receiver pin (RXD) of the converter to the transmitter pin (TXD) of the microcontroller, and vice versa.

6. Identify the Arduino port in the PC's device manager. Open the IDE, navigate to 'tools,' and

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The source code of this project is available for free download at www.electronicsforu.com

select the appropriate board and port.

7. Open a new sketch in the IDE, copy the code, paste it, verify it, and upload it to the ATmega328P micro-controller with a bootloader.

VB6 programming for creating GUI. Set up Visual Studio and create a new window. Add controls and the SerialPort in the toolbox. Write the VB code. Run VB6 on the PC, copy the VB6 program (Fig. 5), paste it into the new project, and create an executable (EXE) file.

Construction and testing

After programming the ATmega328P IC, remove it from the Arduino Uno board. Assemble the circuit and use the programmed IC as shown in Fig. 3. Press F5 to run the VB application. Ensure the correct COM port is specified. Use the Arduino IDE's serial monitor to verify commands and responses, if needed.

Power the hardware, connect the FTDI or Arduino Uno to the PC's USB port, and run the GUI Visual Basic code. The GUI will appear on the screen (Fig. 6). Control the appliances by clicking the GUI buttons. **EFY**



Adeeb Raza is an electronics engineer with about 35 years of practical experience. He is a regular reader and contributor to EFY.